CS610 Solved MCQS 100% Correct

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_____ of TCP/IP layering model, corresponds to basic network hardware.

- · **Physical Layer**
- · Network Interface Layer
- · Internet Layer
- · Transport Layer

_____ protocols of TCP/IP layering model specify how to ensure reliable transfer.

- · Physical Layer
- · Network Interface Layer
- · Internet Layer
- · **Transport Layer**

_____is called an end-to-end protocol because it provide a connection directly from an application on one computer to an application on a remote computer.

- · IP
- · **UDP**
- · TCP

- · None of the given

_____ uses distance vector approach to define routing

- · BGP
- · OSPF
- · **RIP**
- · None of the given

_____ is ideal in a situation where the group is small and all members are attached to contiguous Local Area Networks.

- · **Flood-and -Prune**
- · Configuration-and -Tunneling
- · Core-Based Discovery
- · None of the given

Router that decrements TTL to ___ sends ICMP time exceeded message, with router s address as source address

- · 3
- · 2
- · 1
- · **0**

Protocol addresses are abstractions provided by _____.

- · hardware
- · **software**
- ·

- · internet

Although message exchange can be used to bind addresses, sending a request for each binding is hopelessly inefficient.

- · **True**
- · False

ARP is almost always used to bind a ___-bit IP address to a ___-bit Ethernet address.

- · **32, 48**
- · 24, 32
- · 32, 64
- · 32, 128

In the 1970s large organizations began to acquire multiple networks. Each network in the organization formed island. Employees needed to choose a computer appropriate for each task. So they needed multiple screens, keyboards and computers.

- · False
- · **True**

In which method of Address Resolution Protocol the protocol address is determined by hardware address? Were "T" stands for Table lookup, "C" for Closed-form Computation and "D" for Data Exchange?

- · T
- · D
- · **C**
- · T, C

Which method of Address Resolution Protocol is useful with any hardware?
Were "T" stands for Table lookup, "C" for Closed-form Computation and "D" for Data Exchange?

- · **T**
- · C
- · D
- · C, D

In which method of Address Resolution Protocol the implimentation is more difficult?
Were "T" stands for Table lookup, "C" for Closed-form Computation and "D" for Data Exchange?

- · T, C
- · T
- · C
- · **D**

Which method of Address Resolution Protocol resolution with minimum delay?
Were "T" stands for Table lookup, "C" for Closed-form Computation and "D" for Data Exchange?

- · T, D
- · c
- · T
- · **T, C**

The general form of an IP datagram is with a header followed by data. The header contains information that controls where and how the datagram is to be sent.

- · **True**

- · False

To save traffic, an EGP does not summerize routing information from the autonomous system before passing it to another autonomous system.

- · True
- · **False**

---------- was especially concerned about the lack of high powered computers.

- · IEEE
- · APRA (wrong spelling)
- · EIA
- · **None**

Missing eot indicates sending computer crashed in frame format.

- · **True**
- · False

The ----------term refers to the general concept of a small block of data.

- · **Packets**
- · Data
- · Frame
- · None of given

CRC can detect more errors than a simple checksum.

- · **True**
- · False

The network that uses a ----------, usually consist of a single long cable to which computer attach.

- · Star topology
- · **Bus topology**
- · Ring topology
- · None of the given

LAN that use ATM technology have a ----------

- · Star topology
- · Bus topology
- · Ring topology
- · **None of the given**

A bridge uses ---------- to determine which computer is connected to which segment, and uses the ---------- to determine whether to forward a copy of frame.

- .
- .
- .
- .

---------- has a jitter zero

- · Virtual Private Network
- · **Isochronous Network**
- · Asynchronous Network

- · None of the given

The network with Throughput T and Delay D has a total ---------- bits in transit at a time.

- · D + T
- · D – T
- · **D X T**
- · D / T

One repeater ----------, two repeaters ---------- the maximum cable length limitation.

- · **Double, triple**
- · Double, 4 time
- · half, triple
- · Double, half

End-to-end delivery service is connection oriented.

- · True
- · **False**

A single networking technology is best for all needs.

- · True
- · **False**

Twice NAT allows a site to run servers.

- · **True**
- · False

_____ device is used for multicasting.

- · Hub
- · Switch
- · **Router**
- · none of the given

_____ does not depend on any particular unicast routing protocol.

- · PIM-DM
- · **PIM-SM**
- · PIM-AM
- · none of the given

A routing table contains_____

- · The destination network ID
- · The hop count to reach the network
- · **The router ID of the next hop**
- · All of the given

_____ can be used to propagate information about remote networks.

- · **Dynamic routing**
- · Static routing
- · Address resolution
- · None of the given

_____ protocol is designed to use within an organization.

- · OSPF
- · MEOSPF
- · **MOSPF**
- · none of the given

NAPT stands for _____

- · Network Address and Protocol Translation
- · **Network Address and Port Translation**
- · Network Address and Packet Translation
- · None of the given

In dynamic routing, the routing table is initialized when system boots.

- · **True**
- · False

OSPF includes _____ address mask with each address.

- · 30Bit
- · **32Bit**
- · 34Bit
- · none of the given

Twice NAT fails if an application uses the IP addresses instead of Domain Name.

- · **True**
- · False

_____ uses window mechanism to control the flow of data.

- · IP
- · UDP
- · **TCP**
- · none of the given

TCP uses _____ mechanism to control the flow of data.

- · door
- · **window**
- · acknowledgment
- · retransmission

IGPs stand for _____

- · Internal Gateway Protocols
- · **Interior Gateway Protocols**
- · Intermediate Gateway Protocols
- · None of the given

_____ protocol uses distance vector algorithm.

- · IGP
- · BGP
- · **RIP**
- · none of the given

_____measures distance in network hops, where each network between the source and destination counts as single hop.

- · BGP
- · OSPF
- · **RIP**
- · Non of these

Network Address Translation (NAT) requires device to perform packet translation.

- · **True**
- · False

We use the term _____ to refer to a measure of the path that routing software use when choosing a route.

- · routing path
- · **routing metric**
- · routing
- · switching

Part of the 3-way handshake used to create a connection, requires each end to generate a random _____ sequence number.

- · **32 bit**
- · 16 bit
- · 8 bit
- · 64 bit

Reconstruction of original datagram is called reassembly.

- · **True**
- · False

Reliability is the responsibility of the _____ layer.

- · **Transport**
- · Network
- · Physical
- · Internet

_____ is ideal in a situation where the group is small and all members are attached to contiguous Local Area Networks.

- · **Flood-and -Prune**
- · Configuration-and -Tunneling
- · Core-Based Discovery
- · None of the given

In Direct point to point communication adding the Nth computer requires---------- new connections.

- · None of the given
- · N^2
- · **N-1**
- · $(N^2 - N)/2$

The number of connections needed for N computer in direct point to point communication is equal to:

- · **$(N^2-N)/2$**

- · N(N-1)
- · N^2
- · None of the given

Hardware that calculates a CRC uses two simple components.

- · AND unit and XOR unit
- · **Shift register and XOR unit**
- · Shift register and AND unit
- · None of the given

The Gigabit Ethernet hardware operates at a rate of --------

- · 10 Mbps
- · 100 Mbps
- · **1000 Mbps**
- · None of the given

_____ computes shortest paths in a graph by using weights on edges as a measure of distance.

- · Greedy algorithm
- · Distance vector algorithm
- · **Dijksta's algorithm**
- · Non of the given

Basic LAN technologies such as Ethernet, Token Ring, and FDDI use a _____.

- · **Connectionless service paradigm**

- · Connection-oriented service paradigm
- · Both Connectionless and Connection-oriented service paradigm
- · None of the given

An Internet Address (IP address) is a unique _____ binary number assigned to a host and used for all communication with host

- · 48-bit
- · **32-bit**
- · 24-bit
- · None of the given

The address _____ identifies the physical network to which the computer is attached, while the _____ identifies an individual computer on that network.

- · **prefix , suffix**
- · suffix , prefix
- · suffix , suffix
- · None of the given

_____ places the boundary between the first and second octets

- · **Class A**
- · Class B
- · Class C
- · Class D

_____ places the boundary between the second and third octets.

- · Class A
- · **Class B**
- · Class C
- · Class D

_____ places the boundary between the third and fourth octets.

- · Class A
- · Class B
- · **Class C**
- · Class D

_____ Field of header indicates whether a datagram is a fragment or a complete datagram.

- · **FLAGS**
- · FLAGMENT OFFSET
- · IDENTIFICATION
- · None of the given

_____ provides connectionless service.

- · TCP
- · **UDP**
- · IP
- · None of the given

UDP and TCP are both_____ layer protocols

- · Physical
- · Data link
- · Network
- · **Transport**

_____identifies which application program on receiving computer should receive the data
Logical address

- · Source port
- · **Source port**
- · Destination Port
- · None of the given

_____ identifies the application program that sent the data.

- · Destination Port
- · **Source port**
- · Logical address
- · None of the given

The Border Gateway Protocol (BGP) uses_____ for all communication

- · UDP
- · **TCP**
- · Both UDP and TCP
- · None of the given

Which of the following protocols allows the sender and receiver to enforce polices.

- · RIP
- · OSPF
- · **BGP**
- · RIP and OSPF

ICMP message transport is acted upon by getting ICMP encrypted in IP.

- · True
- · **False**

These ICMP message transport is acted upon _____ as frames on _____

- · Intranet, LAN
- · Internet, WAN
- · Intranet, WAN
- · **Internet, LAN**

Address mask defines how many bits of address are in suffix?

- · True
- · **False**

A computer attached to a given network can only communicate with other computers attached to the same network. Is this a problem with multiple networks?

- · **True**
- · False

The term self-identifying is used for Classful IP addresses because the class of the address can be computed from the address_____.

- · **itself**
- · prefix
- · suffix
- · mask

Find the class of the address:
11110011 10011011 11111011 00001111

- · A
- · C
- · **E**
- · B

There are three standard implementations to improve computational efficiency: Hashing, Direct Indexing and Indirect
Indexing
False
A _____ address-binding table is used for each physical network.
New
Similar
Separate
Old
_____ layer Provides reliable delivery of datagram.

Network
Transport
Datalink
none of the given
Due to revolutionalization of IP-V6 the speed has increased from _____

512 to 1 GB

_____ encapsulates IP datagram as data area in hardware frame.

Network Interface Layer
Datalink Layer
Network Layer
none of the given

_____ Field of header indicates whether a datagram is a fragment or a complete datagram.
FRAGMENT OFFSET

as the Internet grew, the original Classful addressing scheme became a limitation. The IP address space was being exhausted because all networks had to choose one of four possible sizes.

False

As the Internet grew, the original Classful addressing scheme became a limitation. The IP address space was being exhausted because all networks had to choose one of _____ possible sizes.
Three

Header contains all information needed to deliver datagram to the destination computer. But which one of the following is not included:
Rectifier
_____ contains all information needed to deliver datagram to the destination.
Header
one of the parameters, which motivated IP for change is address space. The _____ address space allows for over a million networks. But most networks are class C and too small for many organizations.

32-bit

Class A mask is 255.0.0.0 which is used for _____

Sub netting

Postfix defines how much of address used to identify network.

False
_____ field is used to identify a specific path through the network

FLOW LABEL

_____ is a technique used to Limit datagram size to small MTU of any network

Fragmentation

Hashing is the transformation of a string of characters into a usually shorter fixed-length value or a key that represents the original string.

True

In _____, the protocol address assigned to a computer is chosen carefully so that computer's hardware address can be computed from the protocol address using basic Boolean and arithmetic operations.

Closed-form Computation

Routers use _____ to forward datagram along prearranged path.

Flow label

Mapping between a protocol address and a hardware address is called Address Resolution.

True
Preliminary version of IP was called _____.

IP - Next Generation (IPng)
_____ Source is responsible for fragmentation.

IPV6
End to End delivery Service of IP datagram is _____
Connectionless
Every hardware technology specification includes the definition of the maximum size of the frame data area, which is called the _____ Transmission Unit.
Maximum
IPV6 addresses are _____ bits.
128
MTU Stands for _____
Maximum transmission unit
_____ shows senders preference for low latency, high Reliability.

SERVICE TYPE

NEXT HEADER field in the base header defines type of header it appears at the end of fixed-size base header.

True

There are two standard implementations to improve computational efficiency: Hashing and direct indexing

True

A datagram cannot be larger than _____ of a network over which it is sent.

IP header

The protocol address of the next hop must be _____ to an equivalent hardware address before a packet can be sent.

Translated

The physical addressing in a WAN is similar as in LAN in the way.....

The data is transmitted in packets equivalent to frames.
Each packet has a format with header.
The packet header includes destination and source addresses.
All of the above
In Asynchronous Transfer Mode VPI/VCI fields identify the cells----------
Which one of the following is a method for computing routing table information?

Manual entry
Boot time routing
Static routing
None of the above

All destinations on same switch have same

Router
Information
Next hop
None of the above

Which of the following statement is wrong regarding ATM?
It is a single technology for voice, video and data
It has low jitter and high capacity.
It uses fixed size, small cells, and 48 octet's data
None of the above

Which one of the following is the responsibility of Djikstra' algorithm?

To compute the shortest path between two nodes
To extract next hop information from path information
To insert next hop information into routing table
All of the above

Which of the following statement is true regarding Link-state routing?
Network topology is separated from route computation.
Each switch builds its own routing tables.
Link-state routing uses Djikstra's algorithm.
All of the above

Jitter is significance for ----------

Voice
Video
Data
All of above

The routing table contains information about the .....immediately around it.

Topology of the network
Destination
Both of the above
None of the above

(1) There are three types of Hubs that exist. Which of the following options correctly describes these three?

a. Passive, dormant, special
b. Active , dormant , passive
c. Passive, Active, Turbo
d. Passive, Active, Intelligent

2. A number of different factors determine the cost of a Hub. Which of the following is among these factors?

a. Speed rating
b. Durability
c. Brand
d. Class

Which of the following network topologies among the list below is incorrect?

a. Star

b. Internet
c. Ring
d. Bus

Which of the following is not a feature of Base band Coaxial Cable?

a. Bi-directional
b. Few Kilometer range
c. Proton based
d. Ethernet(basis for 802.3) at 10 Mbps

Which of the following is not a function of a repeater?
Uses Jumper Settings
Transmits data in both directions
No Buffering
Joins two (2) segments of cables
Which level of the TCP/IP reference model routes data/information across a network channel?
Application layer
Data Link Layer
Transport Layer
Network Layer

The function of a modem is to convert signals. Identify the correct signals it converts.
Analogue to Digital
Analogue to Baud rate
Baud rate to Analogue
Digital to Binary

Which of the following is not a LAN – Architecture?.
(a) Protocol architecture
Media access control
Logical Protocol
Logical Link Control
Identify the statement which best describes TCP and UDP.
TCP is a connection oriented Protocol whiles UDP is a datagram service
TCP is a protocol for the Network Layer of the OSI
UDP and TCP could be used interchangeably
TCP is an advanced protocol whiles UDP is a single protocol
Which of the following characteristics best describes a frame?
Data link layer(Header, Trailer, Data)
Network Layer(Header, Trailer, Data)
Transport Layer(Header, Trailer, Data)
Physical Layer(Header, Trailer, Data)

What does the terminology ISDN mean?
Internal Digital Services Network
Integrated Systems Digital Network
Integrated Services Digital Naming System
Integrated Services Digital Network

What does the terminology ATM mean?

Automatic transfer mode
Automatic translation mode
Asynchronous transfer mode
Asynchronous transformation mode

A simple Architecture for File Transfer is made up of :
File transfer application; Communications service module; Network access module

File transfer application; transport layer; Network access module
Network access layer; Transport layer; Application layer
Application layer; Network layer; Session layer
Which of the following statements is true about standards?
Standards create additional cost

Standards help individuals' users to increase effectiveness
Standards allow products from multiple vendors to communicate
Standards cannot freeze technology

Which of the following statement is true about FLOW CONTROL?

Flow control is a function performed by a receiving entity to limit the amount or rate of data on a network.
Flow control adjusts the weights on a network
Flow control is a function performed by the MODEM
Flow control is an advanced form of network management

Identify the most suitable description for ERROR CONTROL?

Error control is an activity that manages deadlock.
Error control is an advanced form of parity bit
Error control is a sub-function of a modem and Hub
Error control is needed to guard against loss or damage of data

What is the full meaning of the terminology SMTP?

State Mail Transfer Protocol
Simple Mail Tailor Protocol

Simple Management Transfer Protocol
Simple Mail Transfer Protocol

Several Computers connected together is called:

Client-server
Client
Computer network
Hub

In which of the following networks does the client submit a task to the server, then the server executes and returns the result for the requesting client station?

(a) Peer-to-Peer
(b) Client-Server
Server-Based
All of the above

In which network topology are all workstations connected in cascade form?
Star
Ring
Mesh
Bus

Which network topology uses a Hub?

Star
Bus
Mesh
Ring
Which of the following topologies is used for Ethernet?

Star
Bus
Ring
All of the above

Computers on a university campus are connected by a/an:
MAN
WAN
LAN
INTERNET

The primary function of the network layer is:

Error detection
Set up a session

Routing
Encryption

Which of the following standards apply to logical link control?

IEEE 802.3
IEEE 802.2
IEEE 802.5
IEEE 802.4

Which organization defines standards for telecommunication?

ITU
IEEE
EIA
ISO

Which of the following statements is a true description of fibre Optic Cable?

Fiber optic cables transfer data in a mesh
Fiber optic cables transfer data in light form
Fiber optic cables are similar to coaxial cables
Non of the above

Which of the following is not true of switching technology?

Packet Switching
Link Switching
Packets are switched in the form of datagram
Circuit switching techniques

Which of the following is a true description of Multi-stage Switching?

More simple control
Increased number of crossed points
More complex control
Reduced Reliability

Flooding in networks and data communication has certain properties. Which of the options below is a property of flooding?

All possible routes are tried
All paths are loaded
All nodes are linked
Can not be used to set up virtual circuit

Identify any of the following statements which truly describe a virtual circuit.

(a) Packets are forwarded more quickly and no routing decisions
(b**) More reliable in functionality**
(c) Call set up phases are present

Packets are centralized at random and more routing decisions are made
Which of the following operating systems software is most suitable for wide area network?

DOS
OS/2
WINDOWS NT
NOVELL NETWARE

Binding is an important function in establishing communication between a Client and a Server. Which of the following statements truly describes binding in protocols?

(a) **Binding protocols to network interfaces specifies that the selected protocol can be sent and received through the selected LAN interface card.**
(b) Binding is a function of the Network Operating Systems Software
Binding is a configuration process in mail server management.
Binding serves as a security layer in operating systems.
(36) In configuring Windows NT 4.0 for Network Access. There are seven (7) areas to be addressed. Identify from the options below the area that should not be addressed.
Protocols
Bindings
IP Address

Autoexec Management

Which of the following is not a network adapter?

3 com LP III

3 com Ether link III ISA

3 com Ether link II Adapter

3 com 3C508 ISA 16 BIT

Which of the following statement is true about an IP address?

IP address is based on your computer network card
IP address is issued by a computer vendor
IP address is a Transport Protocol

IP address is not used on the internet

Which of the following is not associated with the session layer?

Control of dialogue between applications
Dialogue discipline
Data compression
Synchronization

(GOOD LUCK FOR FINAL EXAM)

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